SUK Research and Development Foundation (Incubation Centre)

(a Section 8 Company)

SHIVAJI UNIVERSITY, KOLHAPUR

KISAN-V

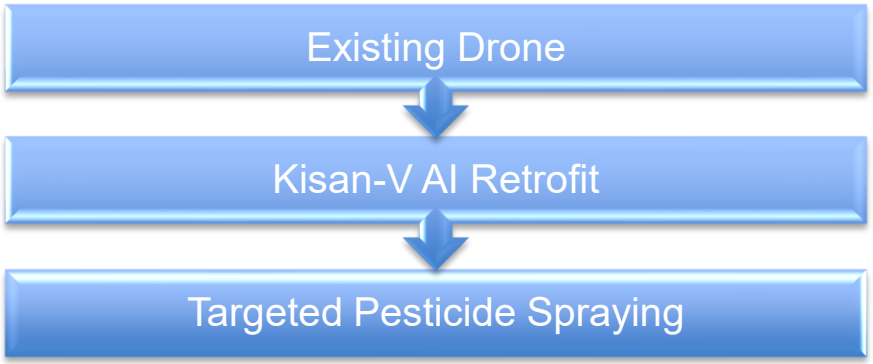
The Indigenous Edge-AI

Presentation by :

Ashish Suryawanshi

Brief Introduction of Idea

KISAN-V is an indigenous, RISC-V based edge-AI retrofit platform that enables real-time “See & Spray” precision pesticide application for agricultural drones.
It upgrades existing drones with on-device intelligence to spray only where weeds or crop stress are detected which is reducing chemical usage, cost, and environmental impact.

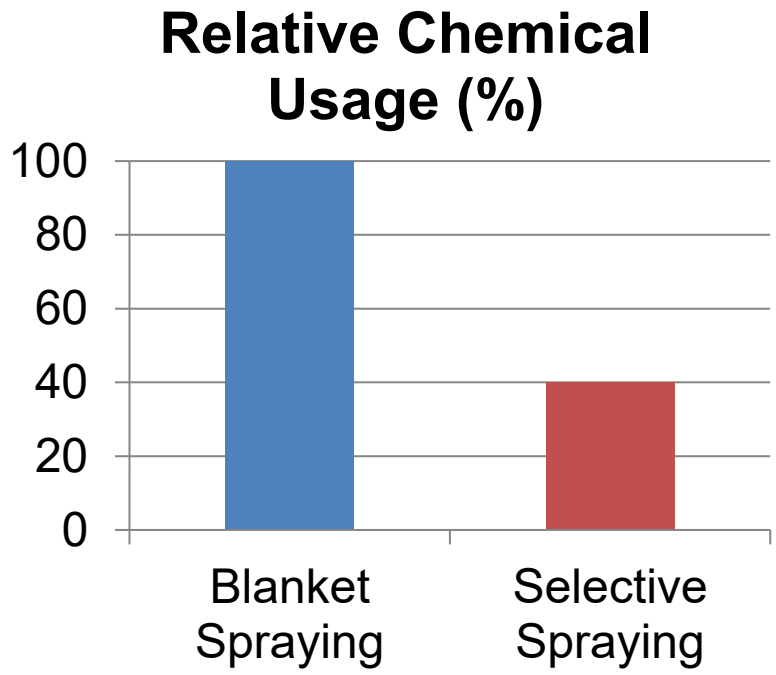


Problem statement

- Pesticide spraying in India is largely **blanket-based**
- Even drones spray **uniformly**, without decision-making

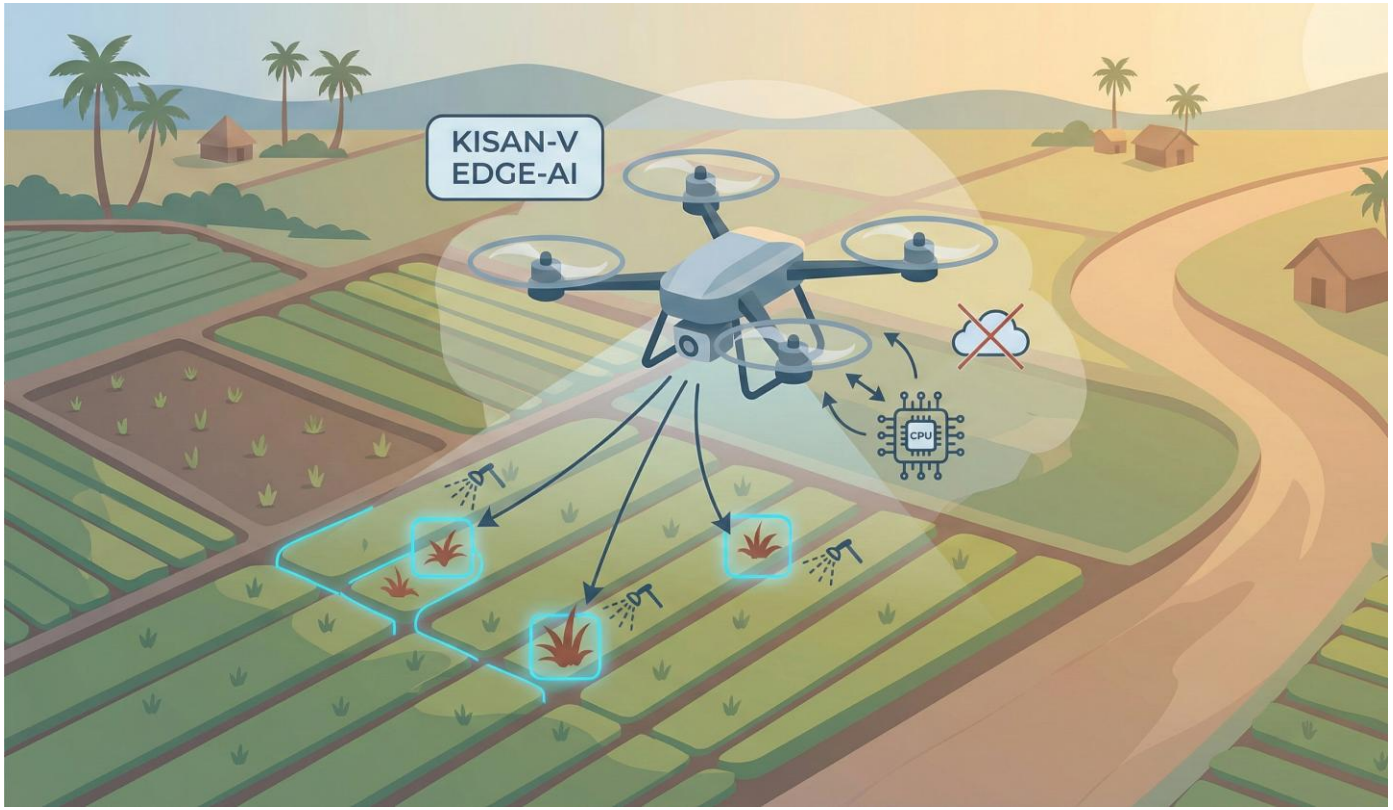
This Leads to:
- Excess chemical usage
- Rising input costs for farmers
- Soil & water contamination
- Health risks

Advanced precision solutions are **too expensive and imported**



Solution of the problem

- **KISAN-V** adds intelligence to existing agricultural drones
- Enables **real-time See & Spray**
- Fully **on-device AI**, no cloud dependency
- Designed for Indian farming conditions



Unique Selling Proposition

Criteria	Traditional Spray Drone	Imported Precision System	Kisan-V
Spraying Method	Blanket / Uniform	Selective (AI-assisted)	**Selective (Real-time See & Spray)**
Intelligence Location	None	Mostly Cloud-based	**Fully On-Device (Edge AI)**
Internet Dependency	No	**Yes**	**No**
Hardware Origin	Imported	Imported	**Indigenous; In house**
Customization for Indian Crops	Very Limited	Limited	**High (Domain-specific)**
Cost Suitability for Small Farmers	Medium	**Expensive**	**Affordable (Retrofit)**
Data Sovereignty	Not Applicable	**Low (Cloud / Foreign servers)**	**High (Local Processing)**
Retrofit Capability	❌	❌	**✓ Yes**
Scalability Across Regions	Medium	Medium	**High**
Alignment with Make in India	❌	❌	**✓ Strong**

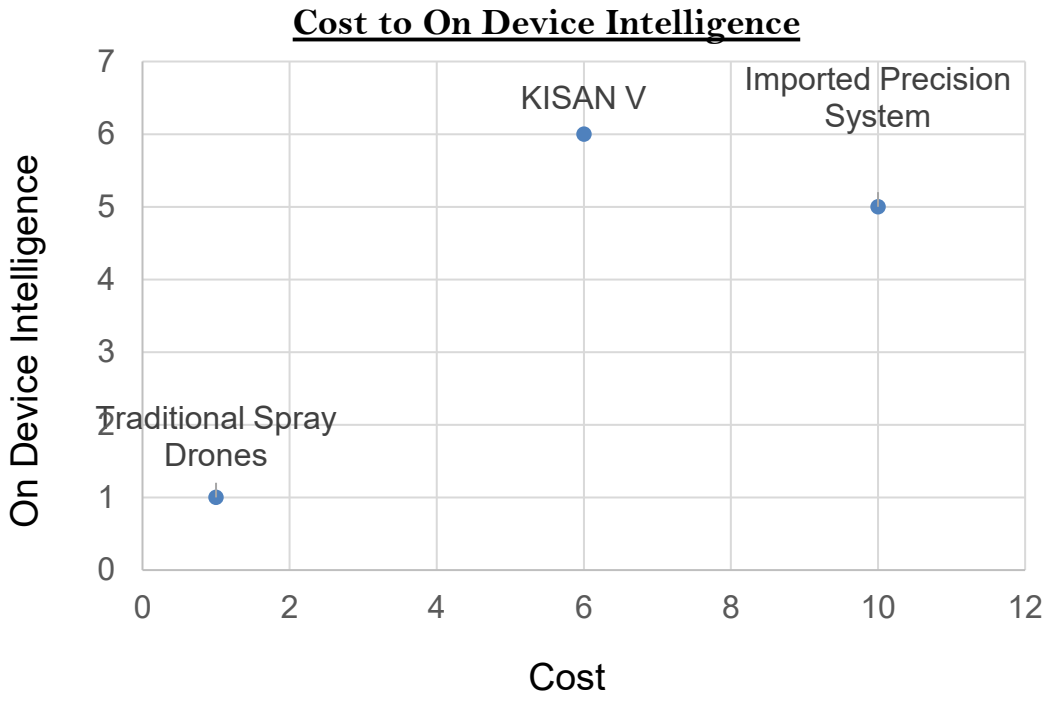
Competition and Barrier To Entry

Competition :

- Imported spraying drones (DJI, etc.)
- Manual & uniform drone spraying
- Cloud-based precision platforms

Barriers to Entry :

- **Custom hardware**
- **On-device AI** optimisation for **Indian farming**
- Domain-specific agriculture datasets
- Deep integration with drone flight stacks



Revenue Model & Customer Turnover

Sr. No.	Description	Value	Notes
Revenue Model			
1.	Hardware Selling Price	₹50,000	1st iteration
2.	BOM + Manufacturing Cost	₹35,000	Prototype estimation
3.	Gross Margin	₹15,000	
Customer Turnover			
1.	Avg acres sprayed / drone / year	1,200 acres	Reference
2.	Avg pesticide cost / acre	₹1,000	Reference
3.	Target Cost saving with KISAN-V	₹600 / acre	Reference
4.	Annual savings per drone	₹7.2 lakh	
5.	Payback period	< 1 season	

Target Market

Primary users:

- Agri-drone service providers
- Agri-tech startups
- Farmer Producer Organisations (FPOs)

End beneficiaries:

- Small & marginal farmers
- Medium and large farms

Geography:

- India (initial)
- Other emerging Agri economies (future)

Proposed Product / Service

Product:

- Plug-and-play **Kisan-V AI Controller Kit** that includes:

  Cameras + Novel Hardware + Mountings and Accessories

Service (Planned):

- AI model updates for different crops
- Maintenance & support
- Customisation and collaboration as HaaS

Milestones

Timeline bullets:

- Concept & feasibility – Completed
- System architecture & research – Completed
- Prototype development – On Going
- Field trials & validation – On Going
- Pilot deployment – Planned

Funding So Far

Description	Funds Invested (₹)	Source of Funds
Hardware Development	1,00,000	Self
Prototype Building	50,000	TADASTITHI GmbH, München
Total	**1,50,000**	—

Seeking Incubation, Grants, And Initial investment upto ₹ 3,00,000

- IP protection planned around hardware-software co-design
- Aligned with DGCA and agricultural drone operational guidelines
- Positioned for support under Digital India, DIR-V, and AgriTech innovation programs

The Team

Thank You

KISAN-V *Indigenous Edge-AI*

See & Spray Retrofit for Precision Agriculture

FOUNDERS :

Ashish Suryawanshi - Co-Founder | Strategy & Business Lead

- Marketing strategy & go-to-market planning
- Budgeting, cost structure & financial forecasting
- Stakeholder coordination

Pratik Suryawanshi - Co-Founder | Technical Lead

- Edge AI Architecture (Hardware & Software Co-Design)
- Realtime systems deployment & optimization
- System integration with drone platforms
- Technical roadmap & prototyping

Open to mentorship, incubation support, and investors.